Huawei Access Node Routing Protocol Flap Troubleshooting SOP for MALL-ACC-03

Node: MALL-ACC-03 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6871

Description

The occurrence of a routing protocol flap on the Huawei Access node MALL-ACC-03 is primarily attributed to the dynamic nature of BGP, OSPF, and ISIS routing protocols, which are integral components of the node's routing infrastructure. The protocols continuously exchange information and adapt to changes in the network topology, causing a flurry of routing updates and adjacency changes. In the event of a flap, the node's routing table may become unstable, resulting in packet loss and network disruptions. This SOP is designed to guide the troubleshooting process for a Huawei Access node experiencing a routing protocol flap, focusing on the ALARM_6871 severity level. Understanding the underlying mechanics of these protocols is essential in identifying the root cause of the issue and implementing an effective resolution. The Huawei Access node's routing protocols are managed by the Huawei proprietary software, which integrates with the node's hardware to provide a robust and scalable routing infrastructure. The protocols operate in the control plane, exchanging routing information and establishing neighbor relationships.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Malleswaram Node.
- Verify the status of redundant links and ensure they are operational.
- Check the node's power supply and ensure it is stable.
- Verify the node's cooling system is functioning correctly.

Potential Root Causes

Cause	Probability
BGP route churn due to changes in the network topology.	High
OSPF adjacency loss due to neighbor relationship issues.	Medium
ISIS LSP flooding due to misconfiguration or hardware issues.	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp process, Expected Output: BGP process information, including the BGP router ID, AS number, and timers., Description: This command displays BGP process information, allowing the troubleshooter to verify the node's BGP configuration and identify potential issues.

Command: display ospf neighbor, Expected Output: OSPF neighbor information, including the neighbor's IP address, state, and timers.,
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Description: This command displays OSPF neighbor information, enabling the troubleshooter to verify the node's OSPF neighbor relationships and identify potential adjacency issues.

Command: display isis lsp flooding, Expected Output: ISIS LSP flooding information, including the LSP identifier, flooding status, and timers., Description: This command displays ISIS LSP flooding information, allowing the troubleshooter to identify potential issues with LSP flooding.
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Resolution Steps

Step 1: Step: Verify and stabilize the node's routing table., Command: clear bgp process, Description: This command clears the BGP process, allowing the node to re-converge and stabilize its routing table.

Step 2: Step: Verify and stabilize the node's OSPF neighbor relationships., Command: clear ospf neighbor, Description: This command clears the OSPF neighbor relationships, allowing the node to re-establish neighbor relationships and stabilize its routing table.

Step 3: Step: Verify and stabilize the node's ISIS LSP flooding., Command: clear isis lsp flooding, Description: This command clears the ISIS LSP flooding, allowing the node to re-converge and stabilize its routing table.

Post-Check Verification

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Command: display bgp process, Expected Output: BGP process information, including the BGP router ID, AS number, and timers., Description: This command displays BGP process information, allowing the troubleshooter to verify the node's BGP configuration and identify potential issues.

Command: display ospf neighbor, Expected Output: OSPF neighbor information, including the neighbor's IP address, state, and timers., Description: This command displays OSPF neighbor information, enabling the troubleshooter to verify the node's OSPF neighbor relationships and identify potential adjacency issues.

Command: display isis lsp flooding, Expected Output: ISIS LSP flooding information, including the LSP identifier, flooding status, and timers., Description: This command displays ISIS LSP flooding information, allowing the troubleshooter to identify potential issues with LSP flooding.
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Roll-back Procedures

- Step: Restore the node's previous BGP configuration., Command: execute bgp configure revert, Description: This command reverts the node's BGP configuration to its previous state, allowing the troubleshooter to roll back changes made during the troubleshooting process.

- Step: Restore the node's previous OSPF configuration., Command: execute ospf configure revert, Description: This command reverts the node's OSPF configuration to its previous state, allowing the troubleshooter to roll back changes made during the troubleshooting process.

- Step: Restore the node's previous ISIS configuration., Command: execute isis configure revert, Description: This command reverts the node's ISIS configuration to its previous state, allowing the troubleshooter to roll back changes made during the troubleshooting process.